SHIVANSH GARG

Jabalpur, India | +91 8602467794 | shivanshgarg2344@gmail.com
linkedin.com/in/shivanshg2344 | github.com/shivansh2344

EDUCATION & CREDENTIALS

Jabalpur Engineering College (JEC) 2023 – 2027 (Expected)
B.Tech in Artificial Intelligence and Data Science | **CGPA: 7.98/10** Jabalpur, India

Kendriya Vidyalaya 2020 & 2022
Class 12th (PCM): **83.8%** | Class 10th: **93.4%** India

National Qualification: GATE Data Science & AI (DA) 2026 – All India Rank (AIR) 3024

EXPERIENCE

Nokia & **BSNL** Oct 2025 – Dec 2025
Advanced AI/ML Telecom Trainee Jabalpur, India

- Spearheaded an AI-driven decision intelligence capstone on 1M+ telecom CDRs and network KPIs, architecting automated ETL pipelines to reduce manual data aggregation time by 20%.
- Engineered an end-to-end churn prediction engine via Scikit-Learn (Random Forest) on imbalanced datasets, achieving 85% classification accuracy.
- Implemented a feature explainability layer mapping root-cause churn drivers (tenure, contract type) to actionable retention strategies, shifting analysis from predictive to prescriptive.
- Developed an interactive what-if scenario simulator allowing dynamic parameter modification to estimate probability shifts, modeling optimized conditions that projected a 30% reduction in customer churn risk.

Google Cloud Apr 2025 – Jul 2025
Generative AI Exchange Program Participant Remote

- Orchestrated 15+ automated prototype workflows to handle 500GB+ of unstructured heterogeneous datasets for LLM fine-tuning and context retrieval.
- Executed 10+ architectural experiments on RAG pipelines, improving information retrieval accuracy by 15% through optimized data transformation and embedding strategies.
- Diagnosed and resolved processing bottlenecks in Generative AI inference workflows, reducing end-to-end pipeline latency by 10%.

TECHNICAL PROJECTS

Multithreaded Deep Packet Inspection (DPI) Engine *C++, libpcap, Multithreading, Posix Threads*

- Engineered a high-performance, multithreaded Deep Packet Inspection engine from scratch in pure C++ to parse L2-L7 network frames (Ethernet, IPv4, TCP/UDP) at wire speed using `libpcap`.
- Architected a concurrent producer-consumer pipeline with custom thread-safe queues, utilizing `std::mutex` and `std::condition_variable` to prevent race conditions and decouple high-speed packet ingestion from protocol parsing.
- Implemented consistent hashing for flow-based load balancing across multiple Fast Path (FP) threads, ensuring accurate stateful connection tracking and microsecond-latency TLS SNI payload extraction.
- Built an O(1) flow-based packet filtering sub-system capable of enforcing access control rules dynamically based on L7 app signatures, seamlessly bypassing encrypted payloads.

VectorForge: A Native Vector DB Inference Engine *C++, Systems Design, Linear Algebra, OOP*

- Built a native, from-scratch vector database and autoregressive transformer inference engine in pure C++ strictly without high-level external ML frameworks.
- Hand-rolled forward-pass neural network layers (Self-Attention, LayerNorm, MLP) and optimized high-dimensional matrix multiplication loops utilizing CPU cache-locality principles for low-latency execution.
- Developed production-grade vector search algorithms (HNSW, KD-Tree) alongside native cosine similarity metrics to navigate 768-dimensional semantic spaces.
- Integrated an efficient tokenization/embedding workflow and exposed CRUD endpoints via a custom REST API, bridging the low-level engine to a fully functioning local RAG pipeline.

OneHealth Integrator Public Health Analytics Platform *Python, Pandas, ETL, Data Orchestration*

- Architected robust ETL pipelines to ingest, normalize, and synthesize highly heterogeneous human, veterinary, and environmental datasets into a centralized analytical data warehouse.
- Engineered automated data harmonization workflows utilizing Pandas and SQL to gracefully handle missing values, timezone disparities, and rigid schema variations across diverse public health sources.
- Designed an interactive public health monitoring layer with real-time alerting systems to track epidemiological trends, providing high-visibility tracking for potential infectious disease vectors.

TECHNICAL SKILLS

- **Languages:** C/C++, Python (Pandas, NumPy, Matplotlib), SQL (CTEs, Window Functions), Java
- **Machine Learning & Data Engineering:** Scikit-Learn, PyTorch, ETL Pipelines, Feature Engineering, Multi-threaded Systems, Concurrency, Memory Management
- **AI Systems:** LLM Inference, RAG Pipelines, Vector Databases (HNSW, KD-Tree), Deep Learning Mathematics
- **Cloud & Developer Tools:** Linux (MSYS2, bash), Google Cloud Platform (Vertex AI, BigQuery), Oracle Cloud (OCI), Git, GitHub, REST APIs

LEADERSHIP & CO-CURRICULAR ACHIEVEMENTS

- **Founding Member, MATRIX JEC:** Scaled the technical society from the ground up, actively mentoring 50+ students through hands-on technical workshops and rigorous coding pipelines.
- **School Captain, Kendriya Vidyalaya:** Directed the student council and managed a student body of 1,000+ peers, successfully coordinating large-scale institutional events and administrative initiatives.
- **Core Team, JEC Tribune & Club Operations:** Facilitated cross-functional collaboration and editorial content oversight for the college's official newsletter, streamlining logistics across multiple student organizations.
- **National Cultural Achievements:** Secured victories in National-Level Competitions for Music and Theatre, demonstrating elite communication, presentation skills, and high-impact execution under pressure.